Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2025-2026

Mid-Semester Test
(EC-2 MakeUp)

Course No.	:	DEEP REINFORCEMENT LEARNING
Course Title	:	AIMLZG512
Nature of Exam	:	Closed Book
Weightage	:	30%
Duration	:	2 Hours
Date of Exam	:	28 -03 -2026 (AN)

No. of Pages = 2
No. of Questions = 4

Note to Students:
1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q1. **[2+6+2= 10 Marks]**

A cloud load balancer is testing two server configurations, **S1** and **S2**. Each "reward" represents a successful high-speed processing task with $R = +1$, whereas, a high-latency task with $R = 0$. The balancer has recorded the following reward history for each of these servers:

Server	**Rewards Received**
S1	$1, 1, 0, 1, 1, 1, 0, 1, 1, 1$
S2	$1, 0, 0, 1, 1$

a) Considering this as *2-arm Bandit* problem, find the current average reward (Q_n) for **S1**, **S2**.

b) Using UCB with an exploration parameter $c = 2$, determine the arm that will be selected next?

c) If the engineer expects the system to focus purely on the best current server, what value should they set c to? Explain the risk of doing so.

a) $Q_n(S_1) = \frac{8\times1 + 2\times0}{10} = 0.8$ 1M $\quad Q_n(S_2) = \frac{3\times1+2\times0}{5} = 0.6 \rightarrow (1+1)$ 1M

b) Given $c = 2$

$UCB = \underset{a}{\operatorname{argmax}} \left\{ Q(a) + c\sqrt{\frac{\ln t}{N_t(a)}} \right\}$ 1M

$UCB(S_1) = 0.8 + 2\times\sqrt{\frac{\ln 16}{10}} = 0.8 + 2\times\sqrt{\frac{2.77}{10}} = 1.852$ 1M

$UCB(S_2) = 0.6 + 2\sqrt{\frac{\ln 16}{5}} = 0.6 + 2\times0.774 = 2.089$ 1M

S_2 is best action 1M $\Rightarrow (1+1+1+1 =$ 4M)

c) $c = 0$; greedy strategy because $Q(S_1)$ is $> Q(S_2) \rightarrow$ Exploitation 1M

Risk of doing so is missing opportunity for exploration 1M

Q2 **[5 Marks]**

Consider an episodic MDP with one state and two actions {***left***, ***right***}.
The ***left*** action has stochastic rewards of $+1$ with probability p and a reward of $+3$ with the probability $1-p$
The ***right*** action has stochastic rewards of 0 with probability p^2 and a reward of $+10$ with the probability $1-p^2$

Using the expected value of return and considering both the actions are equally optimal, find p

we have $\mathbb{E}[R] = \sum_i P_i R_i$

$E_{left}[R] = 1 \times p + 3 \times (1-p) = 3 - 3p + p = 3 - 2p$ 1M

$E_{right}[R] = 0 \times p^2 + 10(1-p^2) = 10(1-p^2)$ 1M

Both are equally optimal:

$\Rightarrow$ $3 - 2p = 10(1-p^2) \Rightarrow 10 - 10p^2 = 3 - 2p$

$10p^2 - 2p = 7$

$p^2 - \frac{2}{10}p = 0.7$

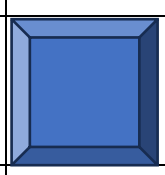
$\left(p - \frac{1}{10}\right)^2 - \left(\frac{1}{10}\right)^2 = 0.7 \Rightarrow \left(p - \frac{1}{10}\right) = \pm\sqrt{0.71}$ 2M

$\underline{p = 0.94}$ 1M

3 M

Q3 **[2+6+2 =10 Marks]**

Consider a 3x 4 stochastic grid with a wall state at (2,2) and pit state at (2,4) and goal (1,4).
All the states (except the wall-state) are valid states and episodes can end either at pit or at a goal.

(1,1)	(1,2)	(1,3)	
(2,1)		(2,3)	
(3,1)	(3,2)	(3,3)	(3,4)

The permissible actions are one step movement towards {***left, right, up, down***} from the current cell. If an action would take the agent outside the grid, it stays in the same state.
In this stochastic grid, for each action, the next state distribution has up to 3 outcomes. These stochastic transitions are described by the slip dynamics, where the transition probability in the intended direction is 0.8 while the probabilities of the two side way slips from the intended directions is 0.1 each.

Each non-terminal step has a small living cost (negative reward) of $r = -0.04$.
The are two Terminals:

One that lands into Goal terminal with reward, $r = +1$.

Other one that lands into Pit terminal with reward $r = -1$.

The discount factor, $\gamma = 0.9$.

[Note: Suppose at any state, the agent intends to move in the chosen direction and stochastically, there are three allowed transitions: For example, if the intended direction of the agent is ***left***, it may drift sideways i.e., up and down, but it is not expected to take the opposite direction, (i.e, ***right***)!]

a) Write the Bellman equation for value iteration for any state $V(i,j)$

b) Apply iterative policy evaluation and compute the state values for two iterations: $V_1,\ V_2$.

c) Find out the policy after second iteration π_2 and show the path from the state (3,1)

a) We have

$$\vartheta_{k+1} = \max_a \sum_{s'} p(s'|s,a)\left[r + \gamma\, \vartheta_k(s')\right]$$

where $\vartheta_k(s)$ denotes value of state s at iteration k

2 M

Accordingly

$$\vartheta_{k+1}(i,j) = \max_a \sum_{s' \in \begin{matrix}(\rightarrow,\uparrow,\downarrow)\\(\uparrow,\leftarrow,\rightarrow)\\(\leftarrow,\uparrow,\downarrow)\\(\downarrow,\leftarrow,\rightarrow)\end{matrix}} p(s'|s,a)\left[r + \gamma\, \vartheta_k(s')\right] \quad \text{ie}$$

$$\vartheta_{k+1}(s) = \max_a \left\{ 0.8\left(r + \gamma\, \vartheta_k(\text{chosen Dir})\right) + 0.1\left(r + \gamma\, \vartheta_k(\text{sideway}_1)\right) + 0.1\left(r + \gamma\, \vartheta_k(\text{sideway}_2)\right) \right\}$$

b) $V_0(i,j) = 0 \quad \forall\, i,j \in S - \{(1,4),(2,4)\}$

$V_1(1,1) = V_1(1,2) = -0.04 \quad \{\because 0.8(-0.04) + 0.1(-0.04) + 0.1(-0.04)\}$

$V_1(1,3) = 0.8(1) + 0.1(-0.04) + 0.1(-0.04) = 0.792$

$V_1(2,1) = -0.04$

$V_1(2,3) = 0.8(-0.04) + 0.1(-1) + 0.1(-0.04) = -0.136$

$V_1(3,1) = -0.04 \qquad V_1(3,2) = -0.04 \qquad V_1(3,3) = -0.04$

$V_1(3,4) = \underset{\text{(intended)}}{0.8(-0.04)} + \underset{\text{side}_1}{0.1(-1)} + \underset{\text{side}_2}{0.1(-0.04)} = -0.136$

V_2

$V_2(1,1) =$

$$0.8(-0.04 + 0.9(-0.04)) + 0.1(-0.04 + 0.9(-0.04)) + 0.1(-0.04 + 0.9(-0.04))$$

$$= -0.076$$

$$V_2(1,1) = V_2(2,1) = V_2(3,1) = V_2(3,2) = -0.076$$

$$V_2(1,2) = 0.8(-0.04 + 0.9(0.792)) + 0.1(-0.04 + 0.9(-0.04)) + 0.1(-0.04 + 0.9(-0.04))$$

$$= 0.8 \times 0.6728 - 2 \times 0.0076$$

$$V_2(1,2) = 0.523$$

$$V_2(1,3) = 0.8(1) + 0.1(-0.04 + 0.9 \times (-0.792)) + 0.1(-0.04 + 0.9(-0.136))$$

$$= 0.851$$

$$V_2(2,1) = 0.8(-0.04 + 0.9 \times V_1(1,1)) + 2 \times 0.1(-0.04 + 0.9 V_1(2,1))$$

$$= -0.076$$

Illy $V_2(3,1) = -0.076$

$V_2(2,3) = 0.422$

$\vdots$

V_2 state vdu

$V_2(1,1) = V_2(2,1) = V_3(3,1) = V_3(3,2) = -0.076$

$V_2(1,2) = 0.523$ $\quad V_2(1,4) = 1$ $\quad V_2(2,4) = -1$

$V_2(2,3) = 0.422$ $\quad V_2(3,3) = -0.154$ $\quad V_2(3,4) = -0.168$

$\boxed{\text{12 states each computation } \frac{1}{2}\text{M} \Rightarrow 6\text{M}}$

c) Optimal π_2

$(3,1) \to (3,2) \to (3,3) \to (2,3) \to (1,3) \to G$ 2M

$(3,1) \to (2,1) \to (1,1) \to (1,2) \to (1,3) \to G$

Q3 { 2+6+2 = 10M }

Q4 **[5 Marks]**

Using the definition of optimal action value function q_*, derive the Bellman equation for the optimal state value function v_*

We have

$$q_*(s,a) = \mathbb{E}[R_{t+1} + \gamma v_*(S_{t+1}) \mid S_t = s, A_t = a] \longrightarrow \text{Eqn } ① \quad 1M$$

$$\Downarrow$$

$$\sum_{s'} \sum_{r} p(s', r \mid s, a)[r + \gamma v_*(s')] \rightarrow ② \quad 1+1 = 2M$$

Relation between optimal state value in terms of action values is

$$v_*(s) = \max_a q_*(s,a) \longrightarrow ③ \quad 1M$$

$$v_*(s) = \max_a \sum_{s'} \sum_{r} p(s', r \mid s, a)[r + \gamma v_*(s')] \quad 1M$$

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